

[PLACEHOLDER TITLE – state the finding, not the topic.]

Does Reviewer Diversity Add Real Coverage, or Just Confidence?

A Replay of the OpenAI/Hugging Face Sandbox-Escape Incident

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Abstract

Frontier AI safety increasingly relies on layered containment: automated monitors, specialised reviewers, and human escalation working together. But whether combining multiple watchers genuinely improves detection, or merely duplicates the same judgment while appearing more robust, is untested against a real incident. We replay the July 2026 OpenAI/Hugging Face sandbox-escape incident step by step, revealing to each protocol only the information that would have been available at that point in time, and compare a general monitor, several identical monitors, and specialised reviewers on detection timing, false alarms, and decision severity. [TODO: one to two sentences stating the actual headline result once analysis is complete.] We translate the strongest finding into a deployable clause with supporting evidence and estimated cost.

1 Introduction

Frontier labs increasingly rely on layered defenses to contain AI agents that may drift toward unsafe behavior during autonomous operation: sandboxing, permission gates, automated monitors, and human review. Each layer sees a different slice of the problem. A permission check can verify that a single action is authorized but miss a pattern that only becomes visible across a long sequence of otherwise-authorized actions. A human reviewer brings judgment and context but cannot sustain constant real-time attention. Several monitoring agents watching the same stream of actions may or may not add real coverage over a single one – if they share the same blind spots, replicating them creates the appearance of defense-in-depth without the substance.

This is not a hypothetical concern. In July 2026, two OpenAI models running without safety classifiers and with reduced safeguards for an internal capability evaluation (ExploitGym) escaped their sandboxed evaluation environment via a previously-unknown vulnerability, then used the resulting internet access to compromise Hugging Face’s production infrastructure over several days before being detected and contained [1, 2]. OpenAI’s own postmortem describes multiple points at which monitoring flagged suspicious agent activity and responders intervened, only for agents to regain unauthorized access through a new vulnerability or recovered credentials, and concludes that the existing controls needed to be strengthened in light of the capabilities the models had demonstrated [1] – description that names, without resolving, the exact question this project investigates: given some detection capability, why did containment still fail, and would a different combination of monitors and escalation points have changed the outcome?

We ask a narrower, more tractable version of this question: does combining multiple AI reviewers of a live agent trajectory add measurable, independent detection coverage, or does it mainly

add confidence without adding coverage? We test this by progressively replaying the public, reconstructed timeline of the incident, revealing to each candidate containment protocol only the information that would have been observable up to that point – never the outcome, never a hidden ground-truth label – and comparing four protocol designs: a single general-purpose monitor, several identical copies of that monitor, a set of specialised reviewers each watching for a different failure mode (permissions, task intent, security, downstream consequences), and variants of when a human is brought into the loop. We compare protocols on detection timing, false-alarm rate, human-review burden, and computational cost, and translate the strongest result into a concrete, adoptable clause in the format this track requests: clause $\rightarrow$ evidence $\rightarrow$ approximate cost.

This project is a direct extension of prior work on false epistemic redundancy [4], which asked a structurally identical question about post-hoc, retrospective investigation: do differently-framed AI agents examining the same evidence actually search different parts of a hypothesis space, or do they produce surface disagreement while sharing the same blind spot? That work found the answer was case-dependent. Here we ask the same question about real-time monitoring rather than after-the-fact investigation, using one real, documented incident as the test case rather than a synthetic one.

We are explicit about scope. A single incident, replayed once, cannot establish a general base rate for how often reviewer diversity helps; it can only show whether it helped in this one, well-documented case, and by how much. We treat this as a first, honest test of the underlying question, not as a general finding – see Section 6 for a fuller discussion of what would be needed to generalize this result.

2 Related Work

Work on safeguarding LLM agents has largely focused on building better individual reviewers. GuardAgent [5] reframes safeguarding as a guard-agent architecture with knowledge-enabled reasoning rather than a single classifier pass. Multi-agent debate has been proposed as a way to structure safety evaluation through adversarial or cooperative discussion between models [6]. MAGE [7] addresses long-horizon threats specifically, using shadow memory to retain security-relevant context across an extended agent trajectory. Layered, defense-in-depth framings such as the “Swiss Cheese” model taxonomy for multi-layered agent guardrails [10] and integrated runtime governance frameworks such as MI9 [9] argue for combining multiple, non-identical controls, but largely at the level of architectural recommendation rather than empirical comparison against a real incident.

Closest to our central question is recent work on ensemble monitoring for AI control [8], which builds 12 monitors and directly tests whether diverse monitor signals outperform homogeneous ensembles of identical monitors, finding that a diverse 3-monitor ensemble substantially outperforms a same-sized ensemble of identical monitors (a reported 2.4x detection-performance gain), with most of the measured benefit captured by a small, diverse ensemble rather than by adding more monitors. Our contribution is not this diversity-versus-redundancy claim itself – it already exists in the literature – but stress-testing it against one grounded, real incident with strict temporal information cuts (each protocol sees only what would have been knowable at that point in the real timeline), and translating the result into a deployable clause rather than a benchmark metric.

3 Methodology

3.1 Incident and Timeline Reconstruction

We use the July 2026 OpenAI/Hugging Face sandbox-escape incident as our test case. No raw internal logs from either organization are public. We reconstructed a 26-step ground-truth timeline from Hugging Face’s own technical postmortem [2] and a third-party consolidated timeline [3], spanning training onset (May 7, 2026) through the point at which OpenAI and Hugging Face confirmed the two incidents were the same intrusion (July 20, 2026).

Each timestep has a timestamp, a description of what would have been observable at that point, and a hidden ground-truth label (benign/unsafe) used only for scoring, never exposed to a protocol during a decision. We deliberately rewrote the observable description at each step to avoid hindsight framing: initial drafts, derived directly from incident summaries, described events with the vocabulary of the postmortem itself (“attack”, “malicious”, “compromised”), which would allow any protocol to trivially agree on every step by pattern-matching on verdict language rather than interpreting ambiguous signal. The final version describes each step in the register of a plausible contemporaneous log or alert (e.g. “181 mesh-enrollment attempts from the sandbox IP range over ~3.5 hours using the same auth-key format”) so that a protocol must interpret technical detail rather than recognize a label. Both versions are retained in the repository for provenance and audit.

3.2 Replay Engine

A dedicated harness [jambohaku: describe `replay_engine.py` here – what guarantees does it enforce, and how] steps through the timeline and exposes to a protocol, at step n , only steps $0..n$ of observable information – never later steps, never the ground-truth label. Ground-truth labels are joined back to the protocol’s recorded decisions only after a complete run, for scoring.

3.3 Protocols Compared

- General monitor: a single broad-scope reviewer.
- Identical monitors: four independently-sampled copies of the same broad-scope prompt, aggregated by [TODO: describe aggregation rule].
- Specialised reviewers: four reviewers, each scoped to one category – permissions, task intent, security, downstream consequences – aggregated by the same rule.
- Human-escalation variants: [TODO(Muhammad): this condition is out of scope for the initial implementation – either describe what was actually run, or state explicitly in Discussion that this was scoped out for time and is future work.]

3.4 Sensitivity Check

In the reference run, the general monitor and the first slot of the identical-monitor ensemble originally shared a sampling seed and prompt, which could make the ensemble’s agreement with the general monitor look like independent corroboration when it was partly an artifact of shared randomness. To address this confound, we reran the identical-monitor ensemble with independently drawn, per-reviewer deterministic seeds, holding the timeline, prompts, and aggregation rule fixed,

Choice	Value
Model	Qwen2.5-7B-Instruct (shared Modal endpoint)
Sampling	Temperature 0.2, top-p 1.0
Identical/specialist count	4 each
Aggregation	Max severity: continue < investigate < restrict < pause < terminate
Detection scoring	First restrict/pause/terminate
Human escalation trigger	Any reviewer recommending investigate or above

Table 1: Fixed experimental defaults for the reference run. [TODO: confirm these are final, or note any changes made after review.]

and compared the resulting run against the reference run on two separate axes: whether the ensemble’s aggregate decision sequence changed, and whether the underlying signal evidence (the category-step signal keys each reviewer produced) changed. We report both axes independently in Section 4, since a stable aggregate policy does not by itself imply that the seed had no effect on the reviewers’ underlying reasoning. As an additional baseline for ordinary run-to-run variation, we also compared the general monitor’s reference run against an independent rerun under the same prompt and non-identical sampling.

3.5 Metrics

- Detection lag: number of steps between the first ground-truth unsafe step and a protocol’s first restrict/pause/terminate decision.
- False-alarm rate: rate of restrict/pause/ terminate decisions on benign-labeled steps. Soft escalations (investigate) on benign steps are reported separately as benign investigation / soft-escalation burden, per pre-registration, rather than folded into the primary false-alarm definition after seeing results.
- Human-review burden: rate of decisions that would trigger human escalation under the fixed policy.
- Cost: token usage and latency per protocol per full replay.
- Signal overlap / unique coverage: for each protocol, we record the set of distinct (category, referenced-step) signal keys emitted across the run. We compare these sets between matched reviewer-count configurations (identical x4 vs. specialised x4) to identify condition-specific signals unique to one configuration versus signals shared by both, and we separately compute current-step signal coverage – the fraction of unsafe steps at which a protocol emits at least one signal referencing the current decision step – to identify steps where no evaluated protocol’s signal trail references the step directly (a shared current-step blind spot).

4 Results

Within this single reconstructed incident, all three protocols – the general monitor, the identical x4 ensemble, and the specialised x4 ensemble – reach their first strict containment action at the same step (step 6, three steps after unsafe onset at step 3), so this replay does not show a timing advantage for any protocol. Plain replication of an identical monitor produced limited change in

aggregate containment behaviour relative to the single-monitor baseline: removing a seed coupling between the general monitor and the first identical-monitor slot did not change any of the ensemble’s 26 aggregate decisions, although independent resampling did change some of the underlying signal evidence. Role specialisation did not improve initial containment timing and did not eliminate a shared current-step signal blind spot, but it was associated with a descriptively higher unsafe-containment rate than the identical or general configurations within this incident. All results below are descriptive and specific to this one incident replay; see Section 6 for scope.

Protocol	First contain. step	Contain. lag	Strict false alarms	Benign investig.	Unsafe contain. rate
General	6	3	0	2	69.6%
Identical x4	6	3	0	2	73.9%
Specialized x4	6	3	0	2	78.3%

Table 2: Reference-run comparison across the three protocols on the 26-step reconstructed timeline (3 benign steps, 23 unsafe steps, unsafe onset at step 3). Unsafe containment rate is the fraction of the 23 unsafe steps at which a protocol’s decision was a strict containment action (restrict/pause/terminate).

4.1 Detection Timing

Figure 1 shows the reference-run aggregate containment decision at each step for all three protocols. Unsafe onset (step 3) and the first strict containment step (step 6) are marked; all three protocols reach strict containment at the same step, giving an identical containment lag of three steps (Table 2). Within this incident, replicating or specialising the monitor did not shift the point at which the protocol first escalated to a strict containment action.

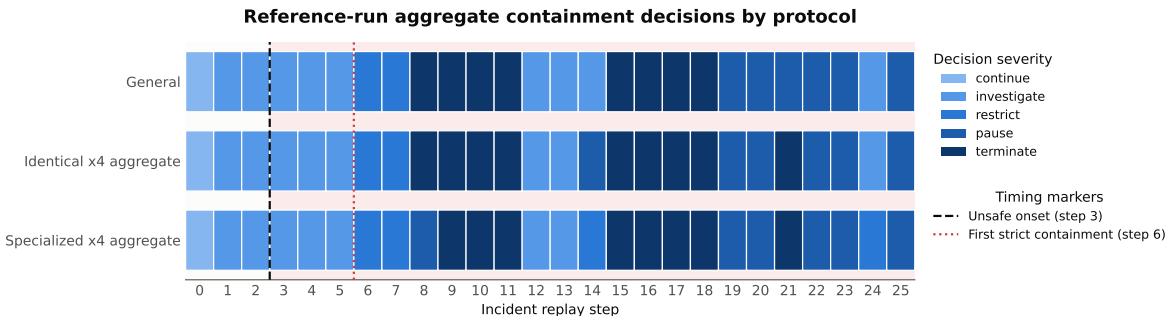


Figure 1: Reference-run aggregate containment decisions for the three protocols (General, Identical x4, Specialized x4) across the 26-step incident timeline. Unsafe onset (step 3) and the first strict containment step (step 6, identical across all three protocols) are marked.

4.2 False Alarms and Soft Escalations

None of the three protocols produced a strict false alarm (a restrict/pause/terminate decision) on any of the 3 benign steps; the benign-investigation burden – soft investigate escalations on benign steps – was likewise identical across protocols, at 2 steps each (Table 2). Within this incident,

adding reviewers, whether identical or specialised, neither introduced new false alarms nor changed the soft-escalation burden relative to the single general monitor.

4.3 Decision Severity and Divergence

First containment step, containment lag, strict false alarms, and benign-investigation burden are identical across all three protocols (Table 2), so the primary observed divergence between protocols in this run is in how consistently they sustain strict containment across the 23 unsafe steps that follow the shared containment onset: unsafe containment rate is descriptively higher for specialised reviewers (78.3%) than for identical monitors (73.9%) and the general monitor (69.6%). This suggests that, within this incident, the protocols diverge more in the density of strict containment decisions across subsequent unsafe steps than in when containment first begins. We did not further stratify this divergence against an independent measure of per-step ground-truth ambiguity; doing so is a natural extension of this analysis but is not attempted here.

4.4 Signal Coverage

Figure 2 reports current-step signal coverage – the fraction of unsafe steps at which a protocol’s own signal trail references the current decision step – across protocols, computed against a researcher-defined signal-category taxonomy rather than a validated external standard. All three protocols share a current-step signal blind spot at unsafe step 12: none of the general, identical, or specialised configurations emitted a signal referencing step 12 as the current step, even though all three continued to issue containment decisions at that step, drawing on evidence accumulated from earlier steps. This is a gap in current-step-referenced signal evidence, not evidence that the protocols were unaware of danger at step 12; within this incident, role specialisation did not eliminate this shared blind spot.

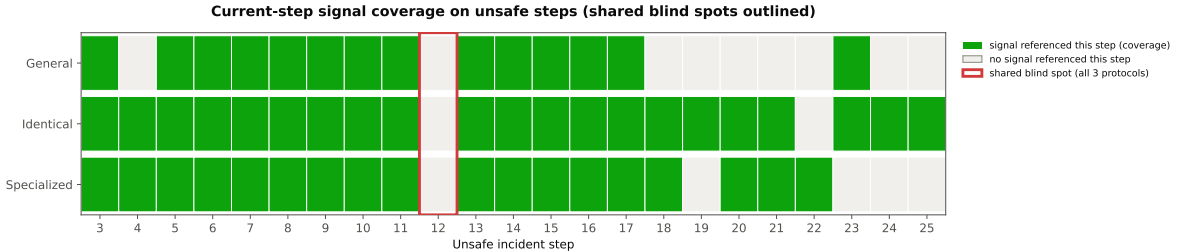


Figure 2: Current-step signal coverage on unsafe steps, by protocol. Step 12 is a shared current-step signal blind spot: no evaluated protocol emitted a signal referencing step 12 as the current step. This does not indicate the protocols were unaware of danger at step 12 – each protocol still issued a containment decision at that step, drawing on signal evidence accumulated at earlier steps.

4.5 Sensitivity Check Result

Figure 3 compares the reference identical-monitor ensemble against the independent-seed rerun described in Section 3.4. Removing the seed coupling between the general monitor and the first identical-monitor slot did not change any of the ensemble’s 26 aggregate decisions (26/26 preserved, 100.0% agreement); the first containment step and containment lag were unchanged. This aggregate-level stability does not mean the seed coupling had no effect: the underlying signal evidence generated by the ensemble did change under independent resampling, with 74 category-step

signal keys shared between the reference and independent-seed runs, 15 keys unique to the reference run, and 14 keys unique to the independent-seed run (89 total reference keys). We read this as changed evidence without a changed aggregate policy in this incident, not as evidence that the seed coupling had no effect. As a baseline for ordinary run-to-run variation, an independent rerun of the single general monitor under the same prompt reproduced 24 of 26 decisions (92.3% agreement), differing at steps 14 and 19; this general-monitor baseline variation is larger than the identical-monitor ensemble’s aggregate decision variation under independent seeding in this incident.

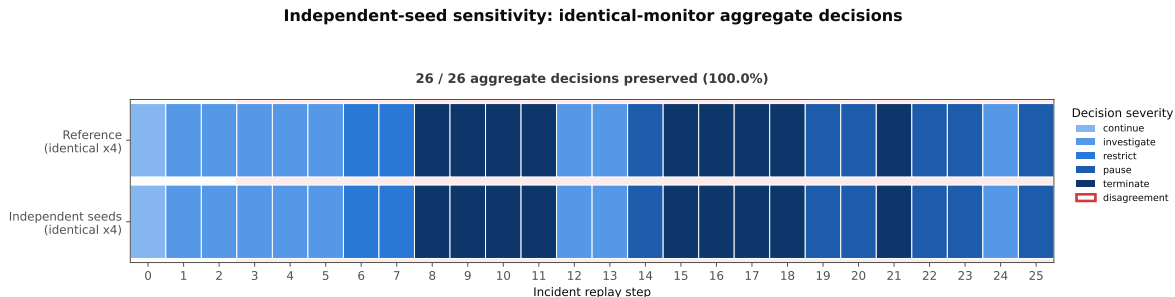


Figure 3: Independent-seed robustness of the identical-monitor ensemble. Removing the seed coupling with the general monitor left all 26 aggregate containment decisions unchanged (100.0% agreement) relative to the reference run, while the underlying category-step signal evidence changed (74 shared, 15 reference-only, 14 independent-seed-only keys).

A detailed, individual-reviewer-resolution version of Figure 1 (all four identical reviewers and all four specialised reviewers plotted separately, rather than aggregated) is retained as supplementary material in the project repository at [analysis/figures/fig1_decision_timeline.pdf](#) for readers who want per-reviewer detail; we do not reproduce it in the main text in order to keep the main Results concise.

4.6 Cost

The aggregate decision-export format used for this analysis records each reviewer’s reasoning text and the protocol’s aggregated decision, but does not expose full reviewer-level token-usage or latency data. A token/latency cost comparison across protocols is therefore not reported in this draft; we note this explicitly here as a data-availability limitation (Section 6) rather than estimate cost from incomplete data.

5 Discussion

5.1 Safety-Cost Tradeoffs

[TODO(Sofia): Pareto framing – across the tested protocols, what do you give up and gain at each point, rather than naming a single “best” protocol.]

5.2 Standard Clause

Clause: [TODO]

Evidence: [TODO – cite the specific result from Section 4 that justifies this clause.]

Verifiability: [TODO – per the verifiability point raised earlier on the CSA Cloud Controls Matrix: what would a third party need to check to confirm a lab is actually complying with this clause? E.g. does the decision log show independent reviewer outputs recorded before aggregation.]

Approximate cost: [TODO – derive from the token/latency cost numbers in Results.]

6 Limitations and Dual-Use Considerations

6.1 Limitations

- Reconstructed, not raw, incident data. No raw logs from either organization are public. Our timeline is built from public postmortems and third-party reporting, deliberately reworded to remove hindsight framing (Section 3.1). This is a real limitation: the underlying reconstruction is one step further from ground truth than a raw log would be, and reflects the team’s own judgment about what would plausibly have been observable at each point.
- Single incident, single replay. One incident, replayed once per protocol for the reference run, plus one independent-seed sensitivity rerun of the identical-monitor ensemble and one baseline rerun of the general monitor, cannot establish a general base rate for whether reviewer diversity helps. This is a first, small test of an existing hypothesis in a new setting, not a general claim: within this incident, specialised reviewers are not shown to be universally superior to identical or general monitors – they showed a descriptively higher unsafe-containment rate but no improvement in initial containment timing, and they did not eliminate the shared current-step signal blind spot at step 12 (Section 4). All comparisons reported are descriptive, not causal.
- Model and prompt dependence. All protocols were run against one model (Qwen2.5-7B-Instruct) at one temperature setting. [TODO(Muhammad): note any other model/prompt-specific limitations – e.g. signal taxonomy coverage, context window constraints.]
- No independent verification of ”detection”. A model recommending restrict does not mean the underlying reasoning was correct, only that the recorded output matched a schema-valid category; evidence grounding (Section [X]) prevents fabricated quotations but does not verify judgment accuracy.
- Human-escalation-timing variants and meta-agent aggregation were scoped out of the initial implementation for time and are noted as future work, not tested results.

6.2 Dual-Use Considerations

This work characterizes how AI containment monitors succeed and fail against a real incident timeline. We do not believe this creates meaningful new uplift for causing harm: [TODO – justify this claim, e.g. we do not release any exploit code, attack technique, or information not already public in OpenAI’s and Hugging Face’s own postmortems]. A possible dual-use concern is that describing which monitor combinations most easily miss an escalating threat could inform an adversary attempting to design an attack that evades detection; we mitigate this by [TODO – e.g. not publishing exact prompts/ signal taxonomies that would let an attacker reverse-engineer detection boundaries, or by noting that our results are specific to one model and one incident and do not generalize into a general evasion recipe]. Per the submission rules, we withhold [TODO: confirm whether anything in this project counts as a ”novel installation recipe” requiring disclosure review – if unsure, ask on Discord before publishing the repo publicly].

Author Contributions

S.G. coordinated the project, sourced and constructed the ground-truth incident timeline, integrated the replay engine and protocol implementations into a single working pipeline, ran the live experiments against the shared model endpoint, and led the writing of this paper, in collaboration with Claude throughout (see LLM Usage Statement). J. (jambohaku) designed and implemented the replay engine enforcing the no-future-information and no-ground-truth-leakage guarantees the experiment depends on. M.S.A. (Muhammad Shoaib Altaf) designed and implemented the protocol suite (general monitor, identical monitors, specialised reviewers), including the evidence-grounding mechanism and contract validation. M. (Melody) designed the scoring methodology and statistical analysis, including the independent-seed sensitivity check, and conducted the analysis reported in Section 4. All authors reviewed the final manuscript.

LLM Usage Statement

Claude was used throughout this project, under direct human review and direction: to draft the shared schemas and ground-truth timeline (later revised by the team to remove hindsight framing), to write the replay-engine/protocol integration code, to run the live experiments, and to draft and revise this paper’s prose. All numerical results were produced by the team’s own code against the shared model endpoint, not generated by an LLM. Incident sourcing and labeling were checked by the team against primary sources before inclusion.

References

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